

AAPL STOCK PERFORMANCE ANALYSIS REPORT

(09 JUNE 2021 – 09 JUNE 2026)

Dataset Overview

This analysis examines the historical performance of Apple Inc. (AAPL) using daily stock market data obtained from Yahoo Finance between **09 June 2021 and 09 June 2026**. The dataset contains approximately **1,370 trading records** and includes Date, Open, High, Low, Close, Adjusted Close, and Volume. The objective is to evaluate stock price behavior, trading activity, returns, and market trends over a five-year period.

Data Transformation Techniques Applied

Several preprocessing and feature-engineering techniques were applied using Python and Pandas. The Date column was converted to datetime format and sorted chronologically for time-series analysis. Daily returns were calculated using percentage changes in closing prices to measure volatility and performance. Additionally, **7-day, 14-day, and 30-day Simple Moving Averages (SMAs)** were generated to identify short, medium, and long-term trends. Monthly returns were computed through data resampling, while trading days were classified as Gain, Loss, or Neutral days to assess market sentiment. Annual performance comparisons were also used to identify the strongest and weakest periods.

Trends Identified

The analysis revealed a strong long-term upward trend in AAPL's stock price. The closing price increased from approximately **\$130 in June 2021** to over **\$310 by June 2026**, representing an overall gain of roughly **138%**. Despite short-term fluctuations, the stock maintained a clear trajectory throughout the analysis period.

Trading volume was highest during **2021–2022**, with several significant spikes exceeding **150 million shares traded**. Although average trading volume declined in later years, stock prices continued to rise, indicating sustained investor confidence.

The moving-average indicators closely followed the price trend and consistently pointed upward, confirming positive momentum. Daily returns were concentrated near zero, with a mean daily return of approximately **0.09%**, indicating that most daily price changes were relatively small. Monthly return analysis showed that **15 of the last 24 months were positive**, with the strongest month delivering approximately **+15%** and the weakest month recording approximately **-8%**.

Features Created

The main analytical features developed during the project included:

- Daily Return (%)
- 7-Day SMA
- 14-Day SMA
- 30-Day SMA
- Monthly Return
- Gain/Loss/Neutral Day Classification
- Best and Worst Year Indicators

These features enabled deeper analysis of stock performance, volatility, momentum, and market sentiment.

Key Insights

The results indicate a positive market bias over the five-year period. Analysis of trading-day outcomes showed that **53.3% of trading days were gain days**, compared to **46.5% loss days** and **0.2% neutral days**. This suggests that positive trading sessions occurred slightly more frequently than negative ones.

Although the average daily return was relatively small, consistent gains compounded over time to produce substantial long-term growth. The strongest performance occurred in **2026**, when AAPL reached its highest observed price level above **\$310 per share**. Furthermore, the daily return distribution suggests manageable volatility, with most trading sessions falling within a moderate return range.

Recommendations

Based on the findings, AAPL demonstrated strong long-term growth characteristics throughout the analysis period. Long-term investors may benefit from focusing on the broader trend rather than short-term fluctuations. Active traders can use moving-average relationships to monitor momentum and identify potential trend continuation opportunities. Future analyses could be enhanced by incorporating technical indicators such as RSI, MACD, and Bollinger Bands, as well as benchmarking AAPL against broader market indices such as the NASDAQ-100 and S&P 500.

Conclusion

Between June 2021 and June 2026, AAPL exhibited strong and sustained growth, increasing from approximately **\$130 to over \$310 per share**. The stock recorded more positive trading days than negative ones, maintained a mean daily return of approximately **0.09%**, and showed predominantly positive monthly performance. Overall, the analysis demonstrates how data transformation, feature engineering, and visualization techniques can be used to uncover meaningful insights from financial time-series data.